Higher National Diploma in Accountancy (New)
First Year, First Semester Examination – 2021
HNDA1123 – Business Mathematics

Instructions for candidates:
Answer five questions **including**
Question number 01
All questions carry **equal** marks.

No. of questions : 06
No. of pages : 04
Time : **Three hours**

Question 01

Write the correct answer.

(i) Natural numbers are also called ..... numbers.

(a) Counting (b) Elementary (c) False (d) Real

(ii) Integers are ..... numbers plus negative numbers.

(a) Irrational (b) whole (c) natural (d) All of the above

(iii) Rational numbers include which type of numbers.

(a) integers (b) natural (c) Whole (d) All of the above

(iv) Which of the following number is natural?

(a) -64 (b) 6.4 (c) $\frac{6}{4}$ (d) 64

(v) The value of $125^{\frac{1}{3}}$

(a) 2 (b) 4 (c) 5 (d) 8

(vi) If $5^{x+3} = 25^{3x-4}$ then the value of x is;

(a) $\frac{5}{11}$ (b) $\frac{11}{5}$ (c) $\frac{3}{11}$ (d) $\frac{11}{3}$

(vii) If $3^{(x-y)} = 27$ and $3^{(x+y)} = 243$ then x is equal to;

(a) 4 (b) 2 (c) 0 (d) 6

(viii) Given that $32\sqrt{2} = 2^a$, the value of a is equal to;

(a) $\frac{11}{2}$ (b) $\frac{2}{11}$ (c) $\frac{12}{2}$ (d) $\frac{11}{3}$

(ix) Simple Interest on Rs. 2500 at 5% per annum for a certain time is 1000. Find the time.

(a) 9 years (b) 7.5 years (c) 8.1 years (d) 8 years

(x) A sum of money at simple interest amounts to Rs. 615 in 3 years and to Rs. 654 in 4 years. The sum is:

(a) Rs.948 (b) Rs.849 (c) Rs.984 (d) Rs.498

(xi) The simple interest on Rs.45 000 in 3 years is Rs.13500.The rate of interest is;

(a) 12 (b) 11 (c) 10 (d) 5

(xii) What is the value of x, $2(-x - 4) = 4x + 16$.

(a) 3 (b) -4 (c) -3 (d) 4

(xiii) A number when added to its half gives 72.The number is ;

(a) 52 (b) 48 (c) 51 (d) 43

(xiv) Find the compound interest on 12000 for 2 years at 10% per annum compounded annually.

(a) 2250 (b) 2525 (c) 2520 (d) 5220

(xv) If a sum of Rs.8000 is lent for 20% per annum at compound interest, the sum of the amount will be Rs.13824 in:

(a) 2 Years (b) 3Years (c) 4Years (d) 1 Year

(xvi) The compound interest on a certain sum of money for 3 years at 20% per annum is Rs. 4368.Find the simple interest at the same rate and for the same time period.

(a) Rs.3600 (b) Rs.2400 (c) Rs.1200 (d) Rs.6300

(Total 2.5 x 16 = 40 Marks)

Question 02

(i) Solve the following equations.

a) $2x + 4y = 14$

$4x - 4y = 4$ (03 Marks)

b) $3x^2 - 9x + 6 = 0$ (03 Marks)

c) $3x + 2y - z = 11$
$2x - 3y + z = 7$
$5x + y - 2z = 12$ (04 Marks)

d) Mr. Kumara invested Rs.300 000; part at 5% simple interest rate, and part at 8% simple interest rate. The total interest on the investment was Rs.21000. How much did he invest at each rate? (05 Marks)

(Total 15 Marks)

Question 03

(i) A firm manufactures a product that is sold for Rs.10 per unit. Variable cost per unit is Rs.6 and fixed cost per period is Rs.1000. Capacity per period is 800 units.

a) Find the revenue and cost functions. Rev = 10 × 800 TC = 6 × 800 (04 Marks)

b) Graph the revenue and cost functions. (04 Marks)

c) Find the number of units sold and the revenue amount (Rs.) at break-even point. (02 Marks)

(ii) Suppose that the demand for soda is Qd = 660–20P, where Qd is the amount of soda that consumers want to buy and *P* is the price of soda. Suppose the supply of soda is Qs = 400+80P, where Qs is the amount of soda that producers will supply.

Find;

a) Equilibrium price (03 Marks)

b) Equilibrium quantity (02 Marks)

(Total 15 Marks)

Question 04

(i) Consider the following data set.

32, 6, 21, 10, 8, 11, 12, 36, 17, 16, 15, 18, 40, 24, 21, 23, 24, 24, 29, 16, 32, 31, 10, 30, 35, 32, 18, 39, 12, 20

Calculate:

a) Range (02 Marks)

b) Mean (02 Marks)

c) Mode (02 Marks)

d) Median (03 Marks)

(ii) Consider the following simple frequency distribution.

X	70	80	90	100	110
f (Frequency)	11	05	03	07	04

Calculate:

a) Mean (02 Marks)

b) Median (02 Marks)

c) Mode (02 Marks)

(Total 15 Marks)

Question 05

A company requires an initial investment of Rs. 140 000. The estimated net cash flows are as follows:

Year	1	2	3	4	5	6
Net cash flow (in Rs.)	18 000	28 000	37 000	37 000	40 000	42 000

Calculate:

i) Payback period (04 Marks)

ii) NPV and comment on it. (Discount rate 10%) (08Marks)

iii) IRR (03 Marks)

(Total 15 Marks)

(Question 06)

i) Mr. Silva plans to set up a fund for his son who will enter the university after 12 years. He thinks that his son will need Rs.800,000 when he enters the university. He finds an account that pays 6% compounded monthly.

a) How much money should Mr. Silva deposit into his account monthly so that his son can get Rs. 800,000 after 12 years? (06 Marks)

b) How much interest will he accrue in his account at the end of the 12 year? (04 Marks)

ii) Suppose you are planning to get a loan amount of Rs.800 000 from a bank at an annual interest rate of 5%. The bank will give you ten years to pay off your loan. Calculate your monthly payment. (05Marks)

(Total 15 Marks)